



3.1 The Chain Rule



Review of Function Compositions

$$(f \circ g)(x) = \underline{\hspace{2cm}}$$

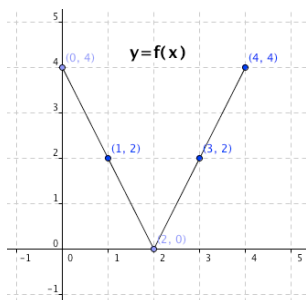
$$(g \circ f)(x) = \underline{\hspace{2cm}}$$

Outer function

$$y = f(g(x))$$

Inner function

Ex 1: Evaluate.



$$g(x) = x - 5$$

x	$h(x)$
-3	1
2	6
4	0
5	2

a) $f(g(6))$	b) $h(f(4))$	c) $f(h(5))$	d) $f(g(h(2)))$
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Ex 2: Identify the outside function, $f(x)$, and inside function, $g(x)$.

a) $y = (3x - 5)^2$	b) $y = \sin 2x$	c) $y = \sqrt{x + 3}$	d) $y = \sin^2 x$
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The Chain Rule

Use the chain rule to differentiate _____ functions.

Ex 3: Circle functions that require the use of chain rule to differentiate.

$$y = 7x^3 + 1$$

$$y = x^2 + 1$$

$$y = \sin 2x$$

$$y = e^{-x}$$

$$y = 3x - 1$$

$$y = x \tan x$$

$$y = \sqrt{x^2 + 1}$$

$$y = (7x + 1)^3$$

$$\frac{d}{dx}[f(g(x))] = \underline{\hspace{2cm}}$$





Ex 4: Find the derivative.

a) $y = (x^2 + 1)^3$	b) $y = \frac{-7}{(2x-3)^2}$
c) $y = \sin 2x$	d) $y = \tan^2 x$
e) $y = \ln(2x)$	f) $y = (\ln x)^3$
g) $y = e^{2x^3-3x}$	h) $y = \frac{1}{e^x}$

Ex 5:

If $f(x) = \sin(\ln(2x))$, then $f'(x) =$

(A) $\frac{\sin(\ln(2x))}{2x}$

(B) $\frac{\cos(\ln(2x))}{x}$

(C) $\frac{\cos(\ln(2x))}{2x}$

(D) $\cos\left(\frac{1}{2x}\right)$



Ex 6:

If $f(x) = e^{2x}(x^3 + 1)$, then $f'(2) =$

- (A) $6e^4$ (B) $21e^4$ (C) $24e^4$ (D) $30e^4$

Ex 7: 2023 AB 5 – part (a)

SCORING STATS!

Guess the mean: _____/9

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

The functions f and g are twice differentiable. The table shown gives values of the functions and their first derivatives at selected values of x .

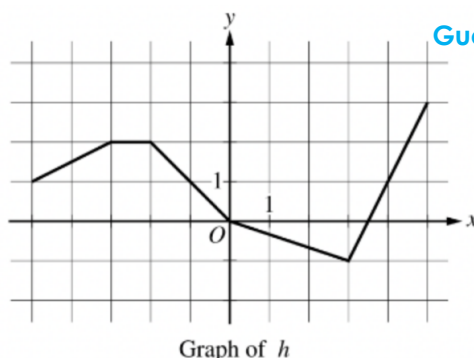
- (a) Let h be the function defined by $h(x) = f(g(x))$. Find $h'(7)$. Show the work that leads to your answer.

Ex 8: 2017 AB6 – parts (b) & (c)

SCORING STATS!

Guess the mean: _____/9

x	$g(x)$	$g'(x)$
-5	10	-3
-4	5	-1
-3	2	4
-2	3	1
-1	1	-2
0	0	-3



Let f be the function defined by $f(x) = \cos(2x) + e^{\sin x}$.

Let g be a differentiable function. The table above gives values of g and its derivative g' at selected values of x .

Let h be the function whose graph, consisting of five line segments, is shown in the figure above.

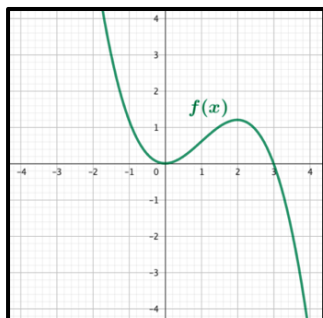
- (b) Let k be the function defined by $k(x) = h(f(x))$. Find $k'(\pi)$.

- (c) Let m be the function defined by $m(x) = g(-2x) \cdot h(x)$. Find $m'(2)$.

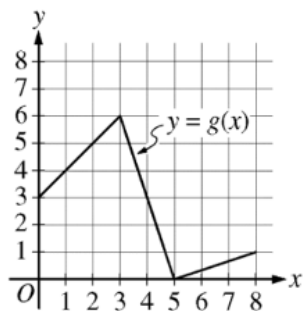
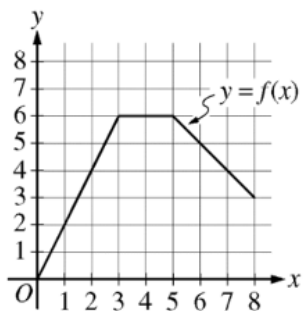


Ex 9: *Medium*

Let $y = f(x^2)$. Use the graph of f given below to determine the sign of $\left. \frac{dy}{dx} \right|_{x=-1}$.



Ex 10: Use the graphs of f and g below to evaluate each expression.



a) *Spicy* 🌶️🌶️

$$\lim_{x \rightarrow 2} \frac{g(3x) - g(6)}{x - 2}$$

b) *Spicy* 🌶️🌶️

$$\lim_{h \rightarrow 0} \frac{f(2(1+h)) - f(2)}{h}$$